CSA1280 – COMPUTER ARCHITECTURE

EXPERIMENT – 32

FULL ADDER USING NAND GATES

AIM: To design and implement a Full Adder using NAND gates circuit using Logisim simulator.

ALGORITHM

1. Open the **Logisim** simulator.
2. Place three input pins: A, B, and Cin.
3. Use NAND gates to construct the **Sum** and **Carry** outputs.
4. Implement the Sum as A ⊕ B ⊕ Cin.
5. Implement the Carry as AB + BCin + ACin.
6. Connect the outputs to Sum (S) and Carry-out (Cout).
7. Test all possible input combinations.
8. Verify the outputs using the truth table.

Truth Table:

A	**B**	**Cin**	**Sum (S)**	**Carry-out (Cout)**
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Sum:

$S = A \oplus B \oplus Cin$

Carry:

Cout = AB + BCin + ACin

RESULT: